AMC 10 Style Practice Test

25 problems · 75 minutes · no calculator · choose one answer (A–E) for each problem

Name: ______________________ Date: ______________________

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1. What is the value of $\frac{2 + 4 + 6 + \ldots + 20}{1 + 3 + 5 + \ldots + 19}$?

(A) $\frac{9}{10}$ **(B)** 1 **(C)** $\frac{11}{10}$ **(D)** $\frac{6}{5}$ **(E)** $\frac{11}{9}$

2. A price first increased by 20%, and then the new price decreased by 25%. How does the final price compare to the original?

(A) 10% lower **(B)** 5% lower **(C)** unchanged **(D)** 5% higher **(E)** 10% higher

3. Anna walked the first 3 km at 6 km/h and the next 3 km at 2 km/h. What was her average speed for the whole walk?

(A) 2.5 km/h **(B)** 3 km/h **(C)** 3.5 km/h **(D)** 4 km/h **(E)** 4.5 km/h

4. A rectangle has perimeter 34 and area 60. What is the length of its diagonal?

(A) 11 **(B)** 12 **(C)** 13 **(D)** 14 **(E)** 15

5. How many two-digit numbers are divisible by 7 but not divisible by 3?

(A) 7 **(B)** 8 **(C)** 9 **(D)** 10 **(E)** 13

6. The average of five numbers is 12. One number is removed, and the average of the remaining four becomes 13. What number was removed?

(A) 6 **(B)** 7 **(C)** 8 **(D)** 9 **(E)** 10

7. How many digits does the number $2^{10} \cdot 5^7$ have?

(A) 7 **(B)** 8 **(C)** 9 **(D)** 10 **(E)** 17

8. A class has 8 students. A team of 3 must be chosen, but two of the students have quarreled and cannot both be on the team. In how many ways can the team be chosen?

(A) 44 **(B)** 48 **(C)** 50 **(D)** 52 **(E)** 56

9. A circle is inscribed in a square with side 8. What is the area of the part of the square lying outside the circle?

(A) $64 - 8\pi$ **(B)** $64 - 16\pi$ **(C)** $64 - 32\pi$ **(D)** $32 - 16\pi$ **(E)** $16\pi - 64$

10. In an arithmetic sequence the first term is 5 and the common difference is 3. Which term of the sequence is the number 302?

(A) 98th **(B)** 99th **(C)** 100th **(D)** 101st **(E)** 102nd

11. A real number x satisfies $x + \frac{1}{x} = 5$. What is $x^3 + \frac{1}{x^3}$?

(A) 110 **(B)** 115 **(C)** 120 **(D)** 125 **(E)** 130

12. Two standard dice are rolled. What is the probability that the sum of the numbers shown is a prime number?

(A) $\frac{1}{3}$ **(B)** $\frac{5}{12}$ **(C)** $\frac{4}{9}$ **(D)** $\frac{1}{2}$ **(E)** $\frac{7}{12}$

13. A right triangle has legs 6 and 8. What is the radius of its inscribed circle?

(A) 1.5 **(B)** 2 **(C)** 2.4 **(D)** 2.5 **(E)** 3

14. What is the least positive integer n such that $n!$ is divisible by 2025?

(A) 9 **(B)** 10 **(C)** 12 **(D)** 15 **(E)** 25

15. How many integers n with $1 \le n \le 100$ are such that $n^2 + n$ is divisible by 6?

(A) 33 **(B)** 50 **(C)** 66 **(D)** 67 **(E)** 83

16. What is the area of a regular hexagon with side 4?

(A) $16\sqrt{3}$ **(B)** $24\sqrt{3}$ **(C)** $32\sqrt{3}$ **(D)** $48\sqrt{3}$ **(E)** 96

17. How many four-digit even numbers have all digits distinct and taken from the set {1, 2, 3, 4, 5}?

(A) 24 **(B)** 36 **(C)** 48 **(D)** 60 **(E)** 72

18. What is the sum of all positive divisors of 360, including 1 and 360 itself?

(A) 1024 **(B)** 1080 **(C)** 1170 **(D)** 1200 **(E)** 1260

19. What is the sum of all real roots of the equation $| \, |x - 2| - 3 \, | = 1$?

(A) 0 **(B)** 4 **(C)** 6 **(D)** 8 **(E)** 10

20. In triangle ABC, point D lies on side BC so that $BD : DC = 1 : 2$. Point E is the midpoint of AD. Line BE meets side AC at point F. What is the ratio $AF : FC$?

(A) 1 : 2 **(B)** 1 : 3 **(C)** 2 : 3 **(D)** 1 : 4 **(E)** 2 : 5

21. A sequence is defined by $a_1 = 1$ and $a_{n+1} = 2a_n + 1$ for all $n \ge 1$. What is a_{10}?

(A) 511 **(B)** 512 **(C)** 1023 **(D)** 1024 **(E)** 2047

22. How many integers x satisfy the inequality $|x - 3| + |x + 2| < 9$?

(A) 6 **(B)** 7 **(C)** 8 **(D)** 9 **(E)** 10

23. What are the last two digits of 3^{2025}?

(A) 01 **(B)** 07 **(C)** 27 **(D)** 43 **(E)** 83

24. Lattice paths from (0, 0) to (5, 5) consist of unit steps right and up. How many such paths never go above the line $y = x$?

(A) 32 **(B)** 42 **(C)** 120 **(D)** 132 **(E)** 252

25. What is the sum of all real numbers x for which $(x^2 - 5x + 5)^{x^2 - 9x + 20} = 1$?

(A) 9 **(B)** 10 **(C)** 13 **(D)** 14 **(E)** 15

Answer Key

Parent page: cut off or do not print for the student. Full solutions are available on the test page online.

1	2	3	4	5
C	**A**	**B**	**C**	**C**
6	7	8	9	10
C	**B**	**C**	**B**	**C**
11	12	13	14	15
A	**B**	**B**	**B**	**C**
16	17	18	19	20
B	**C**	**C**	**D**	**B**
21	22	23	24	25
C	**C**	**D**	**B**	**E**